

## Expt 7 Spectral Estimation

### 7.1 Power Spectral Density Estimation and Spectral Analysis of Noisy Sinusoidal

#### Signals Using Periodogram Method in MATLAB

##### Program:

```
clc; clear; close all

fs=1000; t=(0:999)/fs;

s=sin(2*pi*100*t)+0.5*sin(2*pi*200*t);

n=0.3*randn(size(t));

x=s+n;

figure, plot(x), grid on

title('Signal with Noise')

figure, periodogram(x,[],[],fs)

title('PSD using Periodogram')

figure

X=abs(fft(x));

f=(0:length(X)-1)*fs/length(X);

plot(f(1:500),X(1:500)), grid on

title('Magnitude Spectrum')

figure

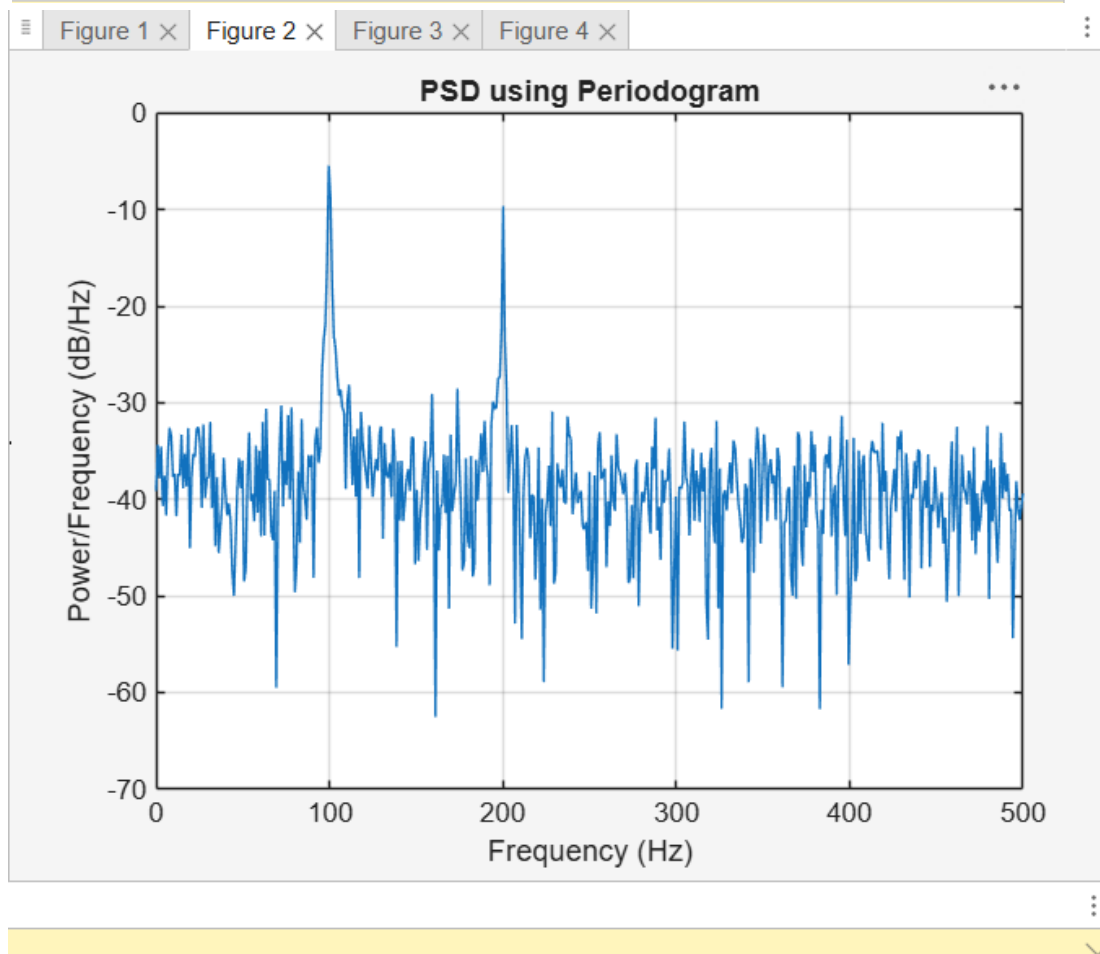
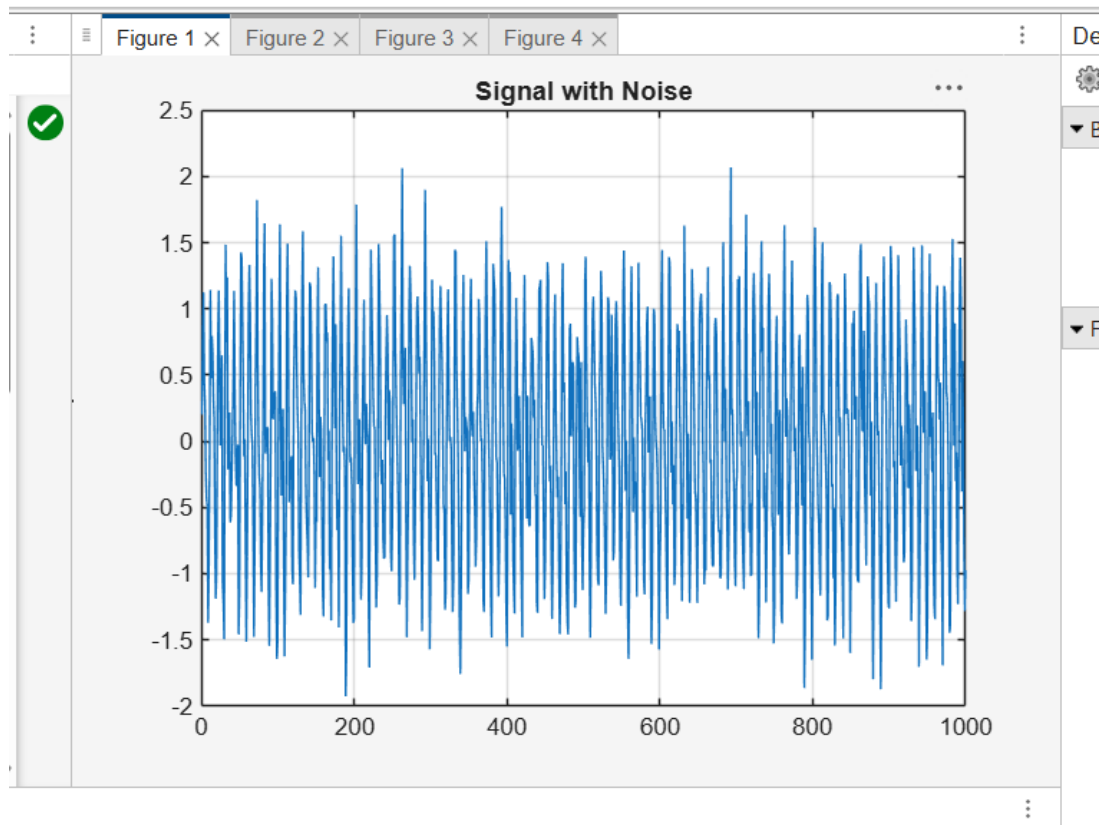
bar([mean(s.^2) var(n)])

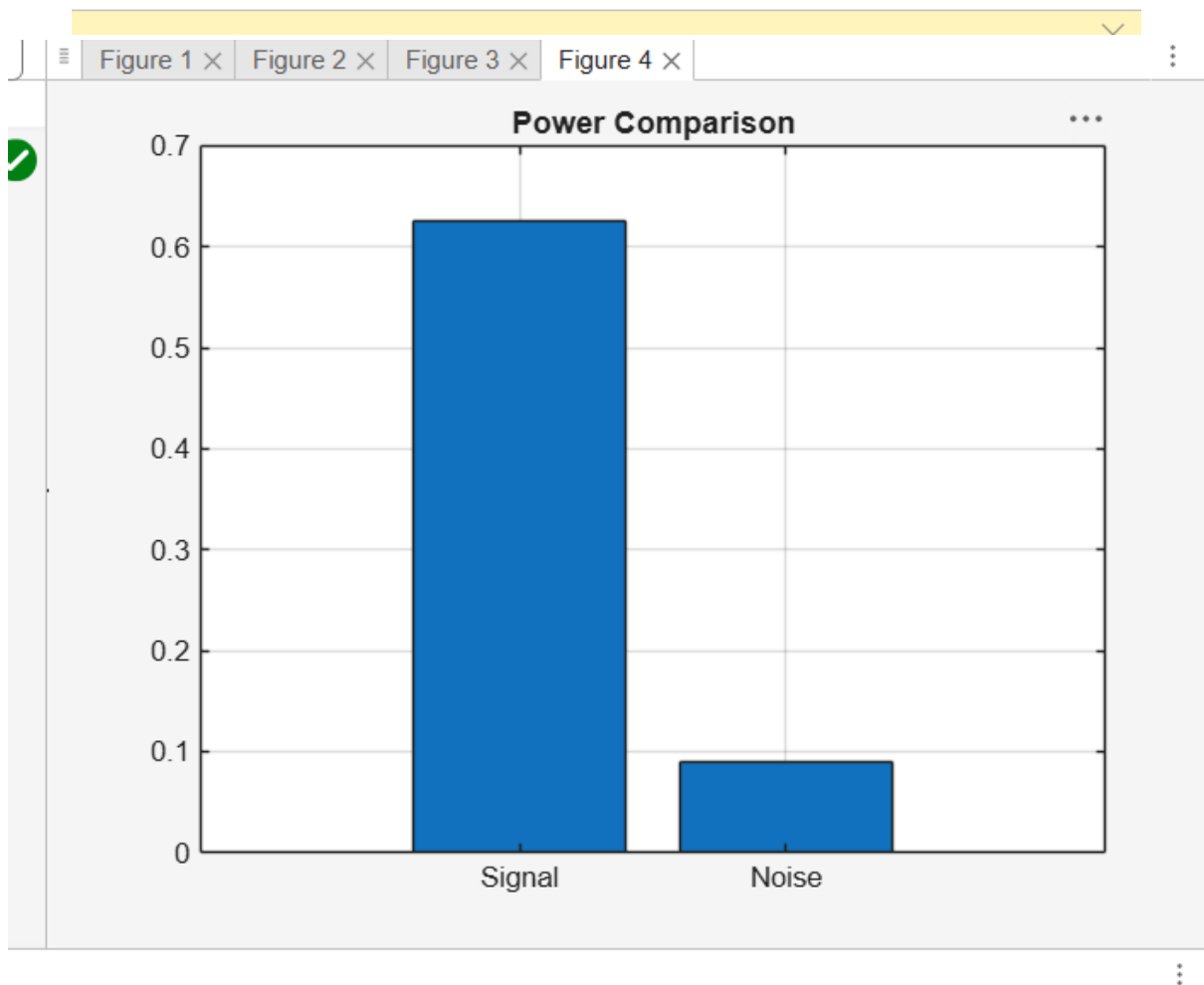
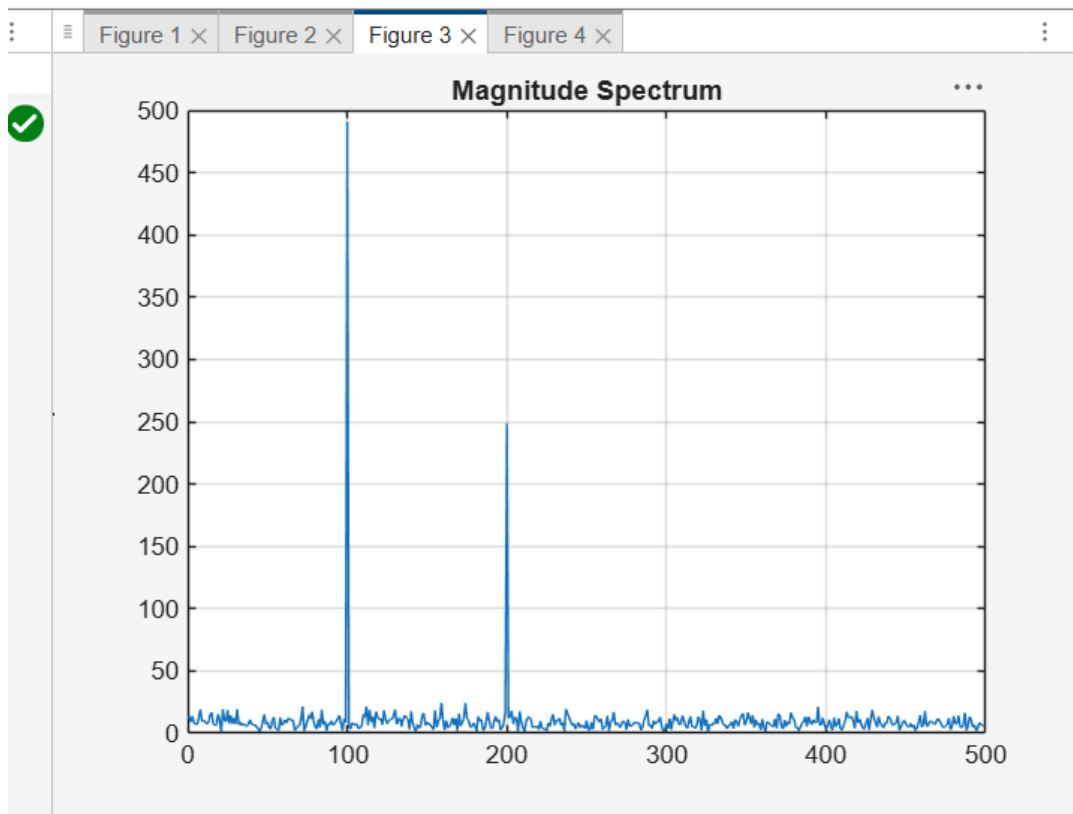
set(gca,'XTickLabel',{'Signal','Noise'})

title('Power Comparison')

grid on
```

##### Output:





## 7.2 Analysis of Windowing Effects on Spectral Estimation Using Rectangular, Hamming, and Hanning Windows in MATLAB

### Program:

```
clc; clear; close all;

% Signal
fs = 1000; N = 256; t = (0:N-1)/fs;
x = sin(2*pi*100*t) + 0.8*sin(2*pi*110*t);
NFFT = 2048; f = (0:NFFT/2-1)*(fs/NFFT);

% Windows
w = {rectwin(N), hamming(N), hann(N)};
names = {'Rectangular','Hamming','Hanning'};

% Processing
for i = 1:3
    xw = x(:).*w{i};
    X = fft(xw,NFFT);
    P{i} = abs(X(1:NFFT/2));
    P{i} = P{i}/max(P{i});
end

%% Individual Spectrums
figure;
for i = 1:3
    subplot(3,1,i);
    plot(f,20*log10(P{i}), 'LineWidth',1.5);
    title(['Spectrum using ' names{i} ' Window']);
    xlim([0 250]); grid on;
end

%% Comparison
figure;
```

```

plot(f,20*log10(P{1}),'k',f,20*log10(P{2}),'r',f,20*log10(P{3}),'b','LineWidth',1.5);
legend(names); title('Window Comparison'); xlim([50 170]); grid on;

%% Window Shapes

figure;

for i = 1:3

    subplot(3,1,i); plot(w{i},'LineWidth',1.5);

    title([names{i} ' Window']); grid on;

end

%% Peak Detection

for i = 1:3

    [pks{i},loc{i}] = findpeaks(P{i},f,'MinPeakHeight',0.3);

    fprintf('\n%s Window Frequencies:\n',names{i});

    disp(loc{i});

end

%% Peak Marking (Hamming)

figure;

plot(f,20*log10(P{2}),'LineWidth',1.5); hold on;

plot(loc{2},20*log10(pks{2}),'ro','MarkerFaceColor','r');

title('Detected Frequencies (Hamming)'); xlim([50 170]); grid on;

%% Main Peaks

fprintf('\nMain Spectral Peaks:\n');

for i = 1:3

    [~,idx] = max(P{i});

    fprintf('%s Window Peak = %.2f Hz\n',names{i},f(idx));

end

```

**Output:**

